

Assignment-5

Section-A

Q1.

→ Loop optimization improves performance by reducing execution time inside loops, since loops execute multiple times.

Techniques

Loop invariant code motion → reduces repeated ^{to} computation

Strength reduction → faster execution

Induction variable elimination → reduces overhead

Loop unrolling → less loop control overhead

Q2.

A directed acyclic graph represents expressions in a basic block.

Purpose:-

Detect common subexpⁿ

Eliminate redundant calculations.

Usefulness

1/1

Eliminates redundancy
Improves optimization
Reduces instruction count.

Q3.

→ Steps to create basic blocks.

1. Identify leaders

- First statement
- Target of jump
- Statement after jump.

2. Group instructions into blocks.

Flow graph

Nodes → basic blocks

Edges → control flow.

Section-B

Q1.

→ 2 3 4 1

Q2.

→ 1 2 3 4

Section-C

Q1 Peep hole optimization

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Optimization technique that examines small instruction sequences and replaces inefficient code.

Eg:-

MOV R₁, R₁ → Remove (redundant)

ADD R₂, 0 → Remove

MUL R₃, 2 → Replace with ADD R₃, R₃

Optimized code

ADD R₃, R₃

Impact:

- Reduces instruction count
- Improves execution speed
- Saves Memory.